Manual

ISE Coursework - Configuration Performance Learning Tool

1. Overview

This manual describes how to use the Configuration Performance Learning (CPL) tool. The tool trains and evaluates three regression models — Linear Regression (baseline), Random Forest (proposed), and Gaussian Process (secondary comparator) — to predict the runtime performance of configurable software systems from a small measured sample. It is implemented in Python using scikit-learn and requires no compilation.

2. Repository Structure

After cloning the repository, the directory layout is as follows:

Path	Description
ise-cpl-coursework/	Repository root
Evaluate.py	Main evaluation script (runs all models on all datasets)
generate_figures.py	Generates Figures 1, 2, and 3 from saved results CSVs
datasets/	CSV performance datasets for all 9 systems (93 workloads)
results/	Output directory — created automatically on first run
requirements.pdf	Dependency list (this file's companion)
manual.pdf	This file
replication.pdf	Step-by-step replication instructions

3. Prerequisites

Ensure the following are installed before running the tool:

- Python 3.9 or higher (Python 3.11 recommended)
- The required Python libraries (see requirements.pdf):
  - **pip install scikit-learn pandas numpy scipy**

The datasets are sourced from the course Lab 2 GitHub repository and must be placed in the datasets/directory. They are already included in the cloned repository.

4. Running the Tool

4.1 Step 1 - Clone the Repository

- **git clone https://github.com/cmpranav17/ise-cpl-coursework.git**
- **cd ise-cpl-coursework**

4.2 Step 2 - Install Dependencies

- **pip install scikit-learn pandas numpy scipy**

4.3 Step 3 - Run the Main Evaluation

The main evaluation script iterates over all 93 dataset–workload combinations, trains all applicable models (LR, RF, and GP where feasible), and saves per-dataset results to results/.

- **python Evaluate.py**

Runtime: approximately 5–15 minutes on a modern laptop, depending on CPU speed. Progress is printed to stdout as each dataset completes.

4.4 Step 4 - Generate Figures

After Evaluate.py completes, run the figure generation script to produce the three plots used in the report:

- **python plot.py**

Figures are saved as PNG files in the results/directory: figure1_mape_bar.png, figure2_pred_vs_actual.png, and figure3_improvement.png.

5. Output Files

After a complete run, the results/ directory contains:

File	Contents
results/all_results.csv	Per-dataset mean MAPE, MAE, RMSE, std, W/T/L counts, Wilcoxon p-values
results/summary_table.csv	Aggregated W/T/L and significance counts (Table 4 in the report)
results/figure1_mape_bar.png	Bar chart of mean MAPE across all 93 datasets (log scale)
results/figure2_pred_vs_actual.png	Predicted vs. actual scatter for 4 representative datasets
results/figure3_improvement.png	Per-dataset relative MAPE improvement + log-log scatter

6. Understanding the Models

The tool evaluates three models on every dataset–workload combination:

Model	Scikit-learn Class	Key Parameters	Evaluated On
Linear Regression (Baseline)	LinearRegression()	None (default)	All 93 datasets
Random Forest (Proposed)	RandomForestRegressor(n_estimators=100, max_features='sqrt')	100 trees, sqrt features at each Split.	All 93 datasets
Gaussian Process (Comparator)	GaussianProcessRegressor(kernel=RBF+WhiteKernel, n_restarts=3)	RBF + WhiteKernel, 3 restarts	lrzip only (< 500 rows)

All models are wrapped in a scikit-learn Pipeline with StandardScaler applied before fitting. Each dataset is evaluated over 30 independent 70/30 train–test splits using different random seeds. The Wilcoxon signed-rank test ($p < 0.05$) is used to determine statistical significance.

7. Troubleshooting

Issue	Likely Cause	Fix
ModuleNotFoundError	Missing dependency	Run: pip install scikit-learn pandas numpy scipy
FileNotFoundError on CSV	Datasets folder missing or mis-named	Ensure datasets/ folder is present in the repo root (it is included in the s
GP evaluation hangs	Large dataset fed to GP	Expected GP is automatically skipped for dataset with >= 500 row
Results CSV already exists	Re-running after partial run	Delete results/ folder or let it overwrite Evaluate.py overwrites existing